# Chapter 36: The Short-Run Trade-off between Inflation and Unemployment

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If policymakers could temporarily choose between higher inflation or higher unemployment, which would you prioritize—and what hidden costs might your choice create for the future?

# Learning Objectives

- 36.1 Analyze the short-run trade-off between inflation and unemployment using the Phillips curve and explain why no permanent trade-off exists.
- 36.2 Derive and interpret both the short-run and long-run Phillips curves using scenarios, graphs, or data, including the role of the natural rate of unemployment.
- 36.3 Explain how adaptive and rational expectations shape inflation dynamics and influence the position and slope of the Phillips curve.
- 36.4 Evaluate how changes in monetary policy affect inflation, unemployment, and the economy's transition between short-run and long-run equilibria.
- 36.5 Calculate and interpret the sacrifice ratio in the context of disinflation, output changes, and historical movements of the Phillips curve.

# Key Points

- The Phillips curve shows a short-run negative relationship between inflation and unemployment: expanding aggregate demand lowers unemployment but raises inflation, while contracting demand raises unemployment but reduces inflation.
- The trade-off exists only in the short run. As expected inflation adjusts, the short-run Phillips curve shifts, leaving a vertical long-run Phillips curve at the natural rate of unemployment.
- Short-run Phillips curves also shift with supply shocks. An adverse supply shock (e.g., a spike in oil prices) makes the inflation–unemployment trade-off worse, raising inflation for any unemployment rate—or unemployment for any inflation rate.
- Reducing inflation through slower money growth moves the economy up the short-run Phillips curve and temporarily raises unemployment. The cost of disinflation depends on how fast inflation expectations adjust; credible anti-inflation policy can lower this cost by speeding up expectation adjustment.

# Introduction

- In the long run, inflation & unemployment are unrelated:
  - The inflation rate depends mainly on growth in the money supply.
  - Unemployment (the "natural rate") depends on the minimum wage, the market power of unions, efficiency wages, and the process of job search.

- One of the Ten Principles:
  ***In the short run, society faces a trade-off between inflation and unemployment.***

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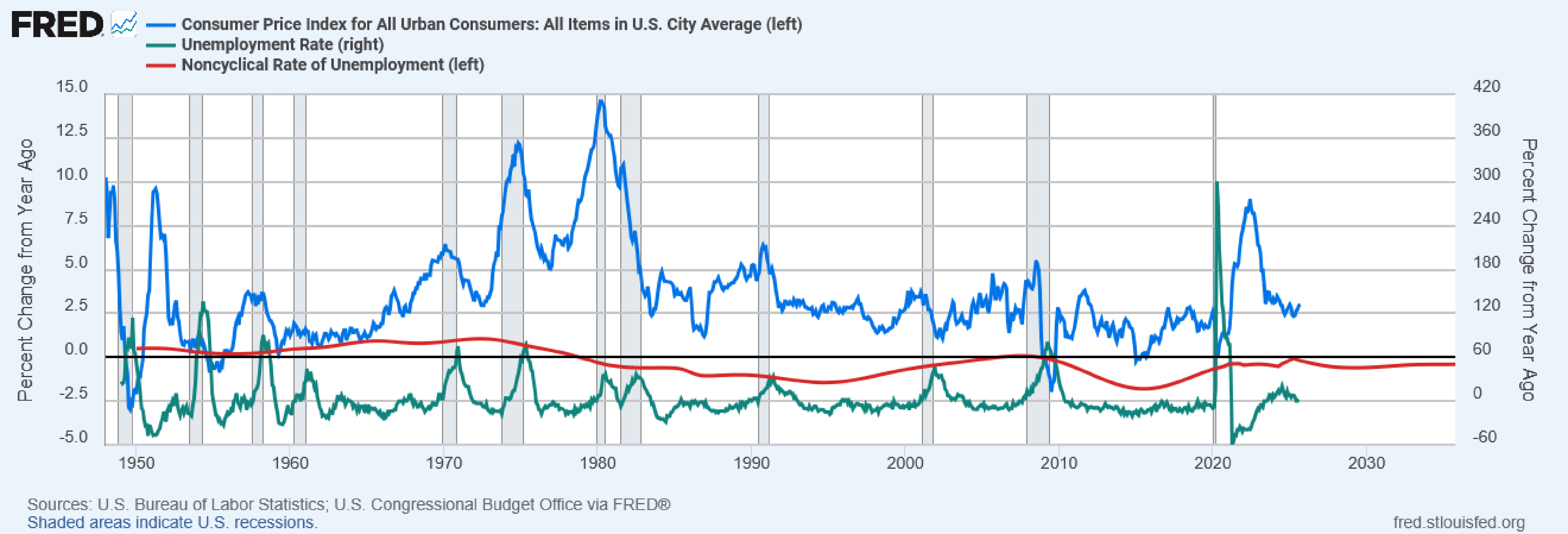
# A Quick Look at the Data

- In the long run, inflation & unemployment are unrelated:

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# The Phillips Curve

- **Phillips curve**: shows the short-run trade-off between inflation and unemployment

- 1958: A.W. Phillips showed that nominal wage growth was negatively correlated with unemployment in the U.K.

- 1960: Paul Samuelson & Robert Solow found a negative correlation between U.S. inflation & unemployment, named it "the Phillips Curve."

# Deriving the Phillips Curve

Suppose ***P*** = 100 this year.

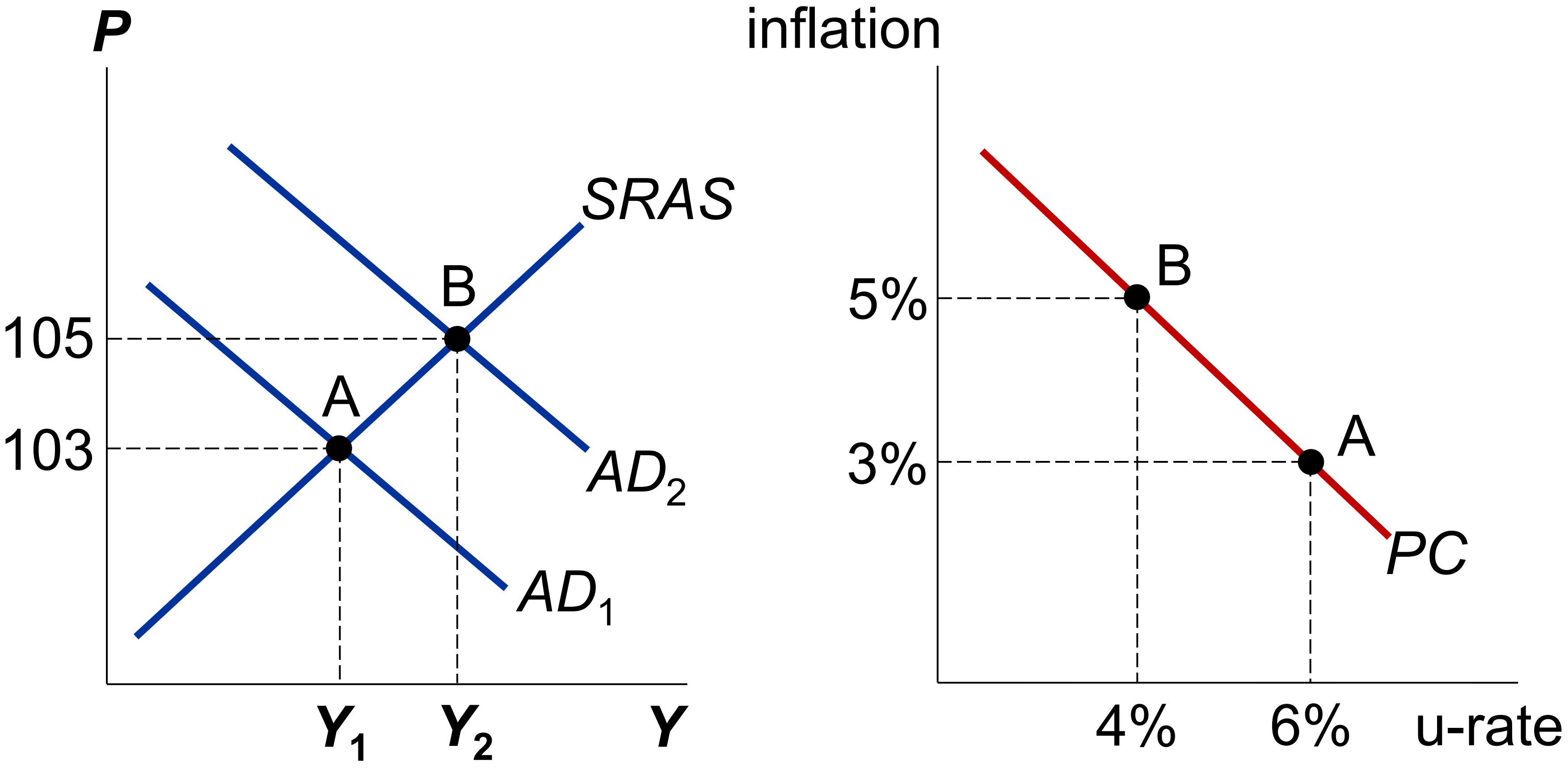
The following graphs show two possible outcomes for next year:

A. Aggregate demand low, small increase in ***P*** (i.e., low inflation), low output, high unemployment.

B. Aggregate demand high, big increase in ***P*** (i.e., high inflation), high output, low unemployment.

# Deriving the Phillips Curve

**A.** Low aggregate demand, low inflation, high u-rate




**B.** High aggregate demand, high inflation, low u-rate

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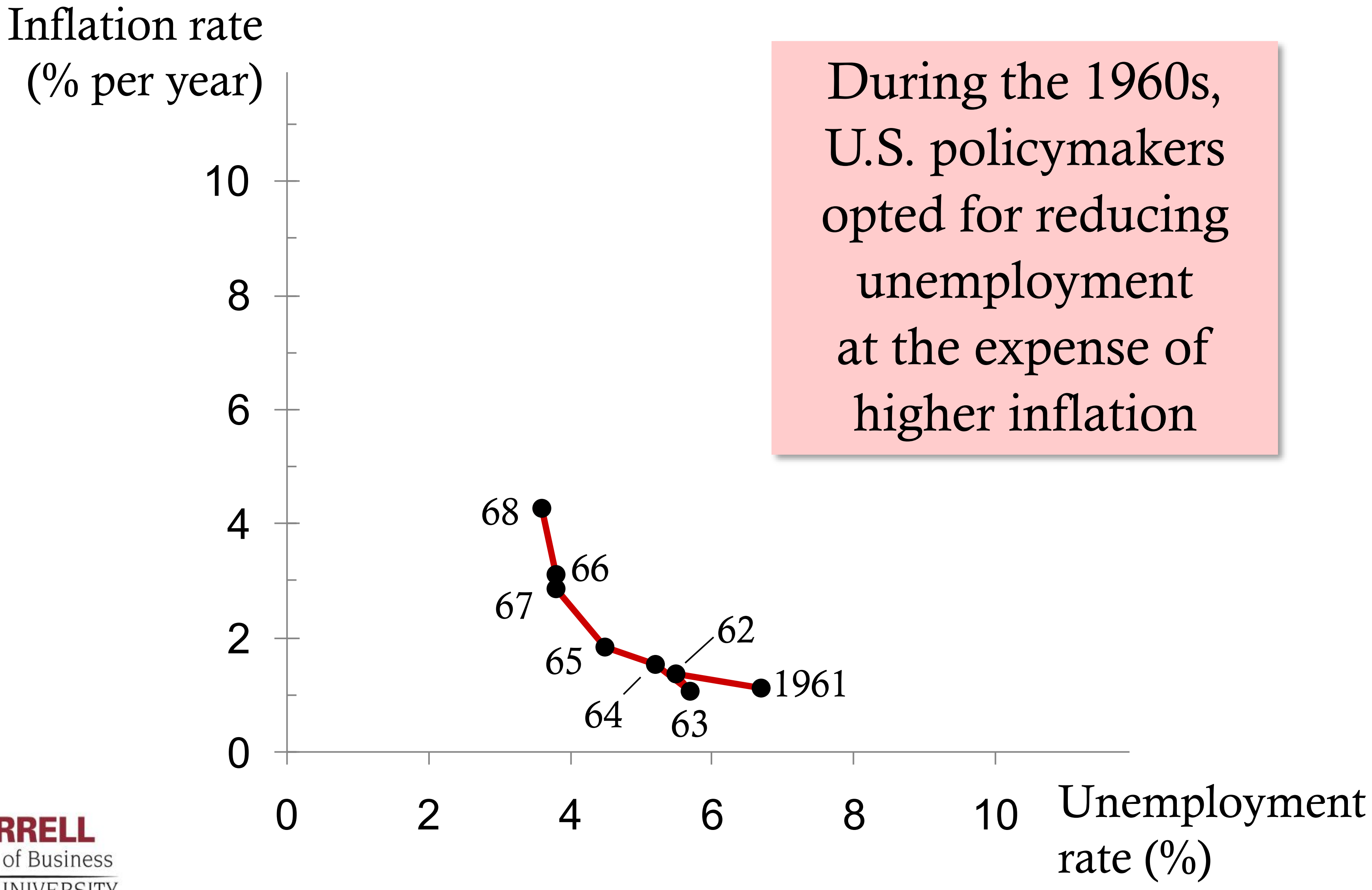
# The Phillips Curve: A Policy Menu?

- Since fiscal and monetary policy affect aggregate demand, the *PC* appeared to offer policymakers a menu of choices:
    - low unemployment with high inflation
    - low inflation with high unemployment
    - anything in between

- 1960s: U.S. data supported the Phillips curve. Many believed the *PC* was stable and reliable.

# Evidence for the Phillips Curve?




During the 1960s, U.S. policymakers opted for reducing unemployment at the expense of higher inflation

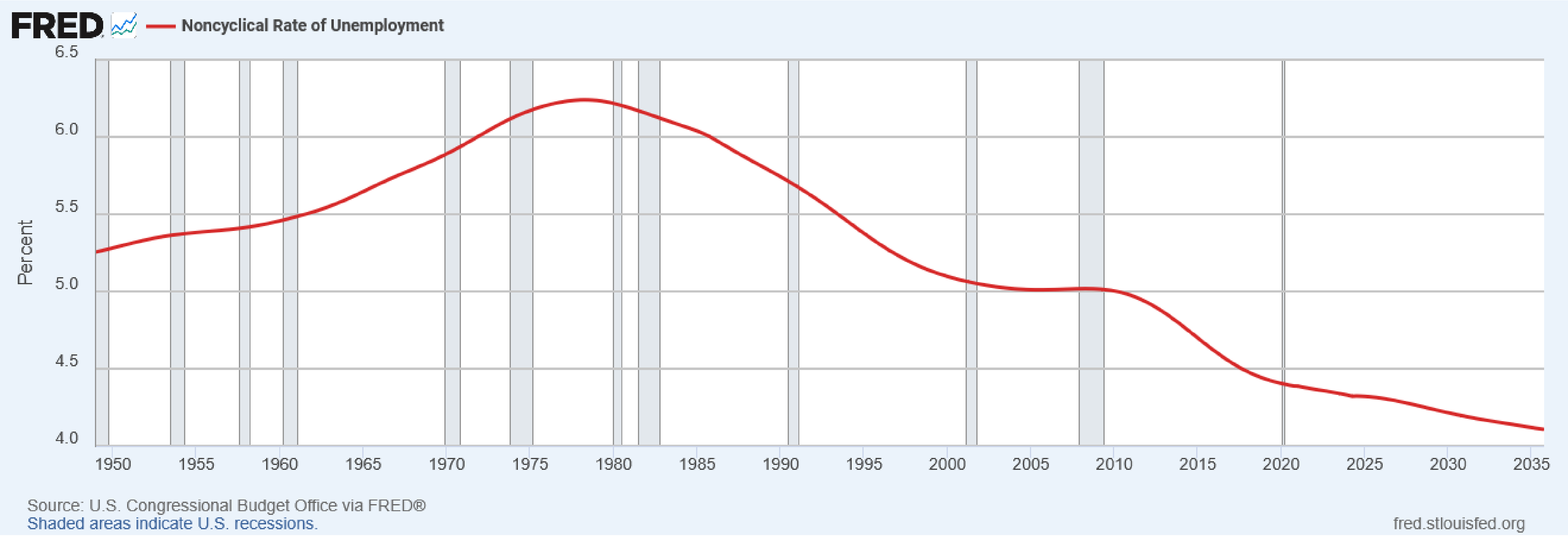
# The Vertical Long Run Phillips Curve

- 1968: Milton Friedman and Edmund Phelps argued that the tradeoff was temporary.

- **Natural-rate hypothesis**: the claim that unemployment eventually returns to its normal or "natural" rate, regardless of the inflation rate

- Based on the classical dichotomy and the vertical *LRAS* curve

# Noncyclical Rate of Unemployment

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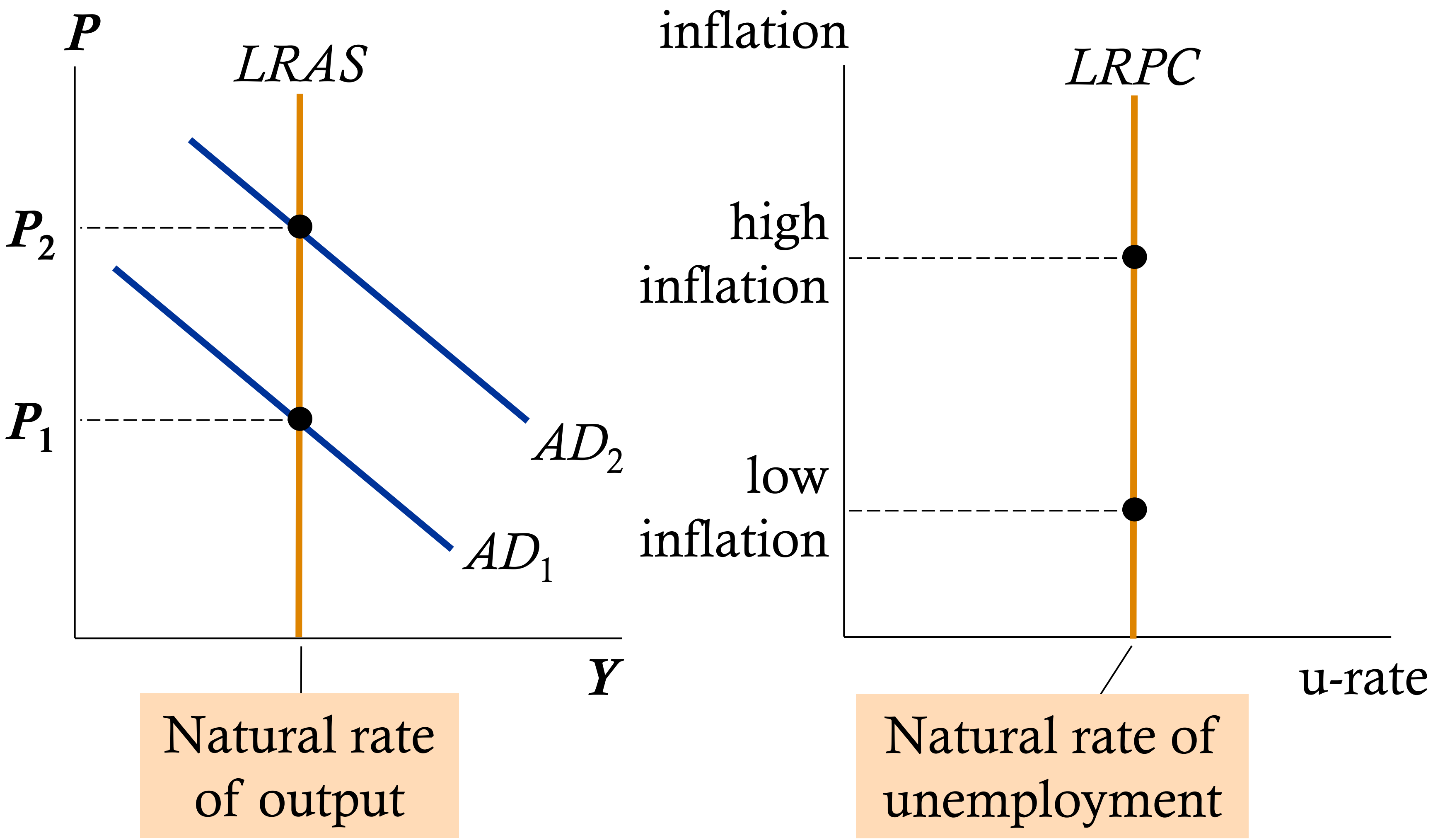
# The Vertical Long Run Phillips Curve

In the long run, faster money growth only causes faster inflation.



# Reconciling Theory and Evidence

- Evidence (from ’60s): *PC* slopes downward.

- Theory (Friedman and Phelps): *PC* is vertical in the long run.

- To bridge the gap between theory and evidence, Friedman and Phelps introduced a new variable: **expected inflation** – a measure of how much people expect the price level to change.

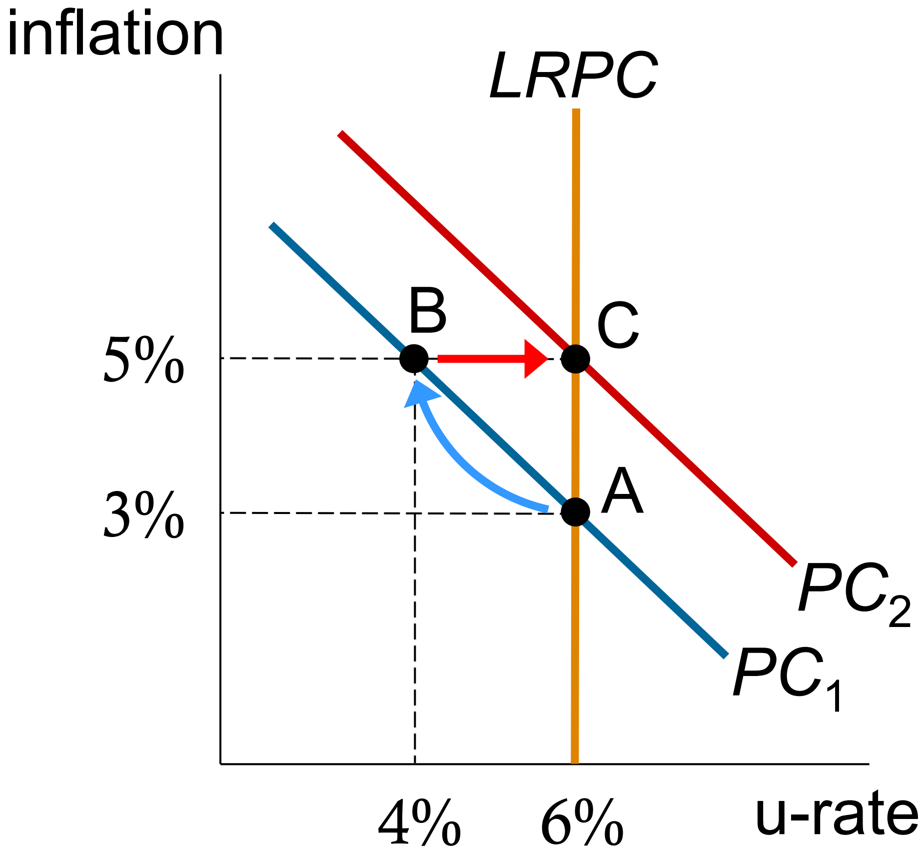
# The Phillips Curve Equation

$$\begin{matrix}\text{Unemp.} \\ \text{rate}\end{matrix} = \begin{matrix}\text{Natural} \\ \text{rate of} \\ \text{unemp.}\end{matrix} - \boldsymbol{a}\left(\begin{matrix}\text{Actual} \\ \text{inflation}\end{matrix} - \begin{matrix}\text{Expected} \\ \text{inflation}\end{matrix}\right)$$

- **Short run**
  - Fed can reduce u-rate below the natural u-rate by making inflation greater than expected.
- **Long run**
  - Expectations catch up to reality, u-rate goes back to natural u-rate whether inflation is high or low.

# How Expected Inflation Shifts the *PC*

- Initially, expected & actual inflation = 3%, unemployment = natural rate (6%).
- Fed makes inflation 2% higher than expected, u-rate falls to 4%.
- In the long run, expected inflation increases to 5%, *PC* shifts upward, unemployment returns to its natural rate.

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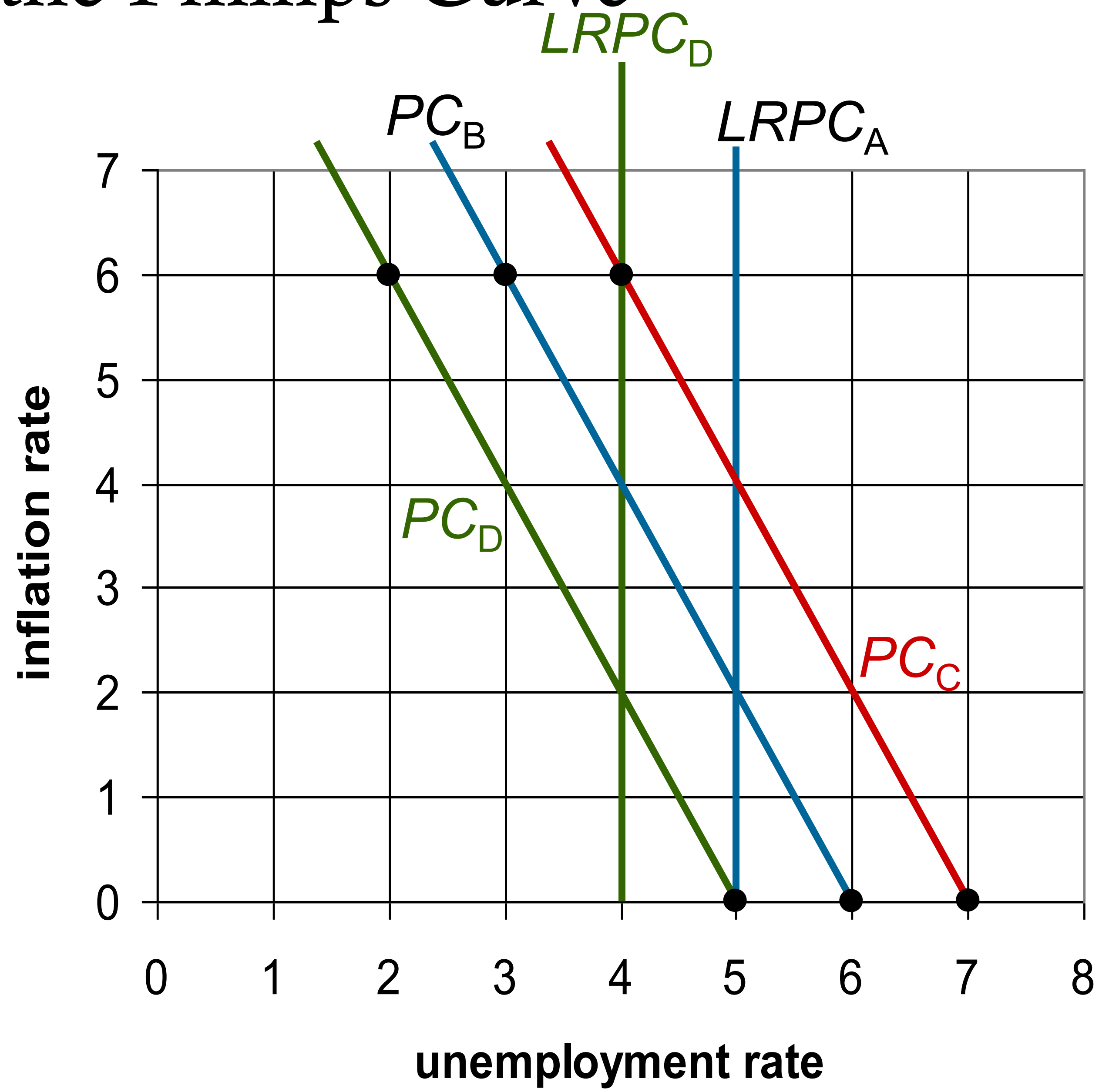
# Activity: Shifting the Phillips Curve

Natural rate of unemployment = 5%
Expected inflation = 2%
In *PC* equation, ***a*** = 0.5

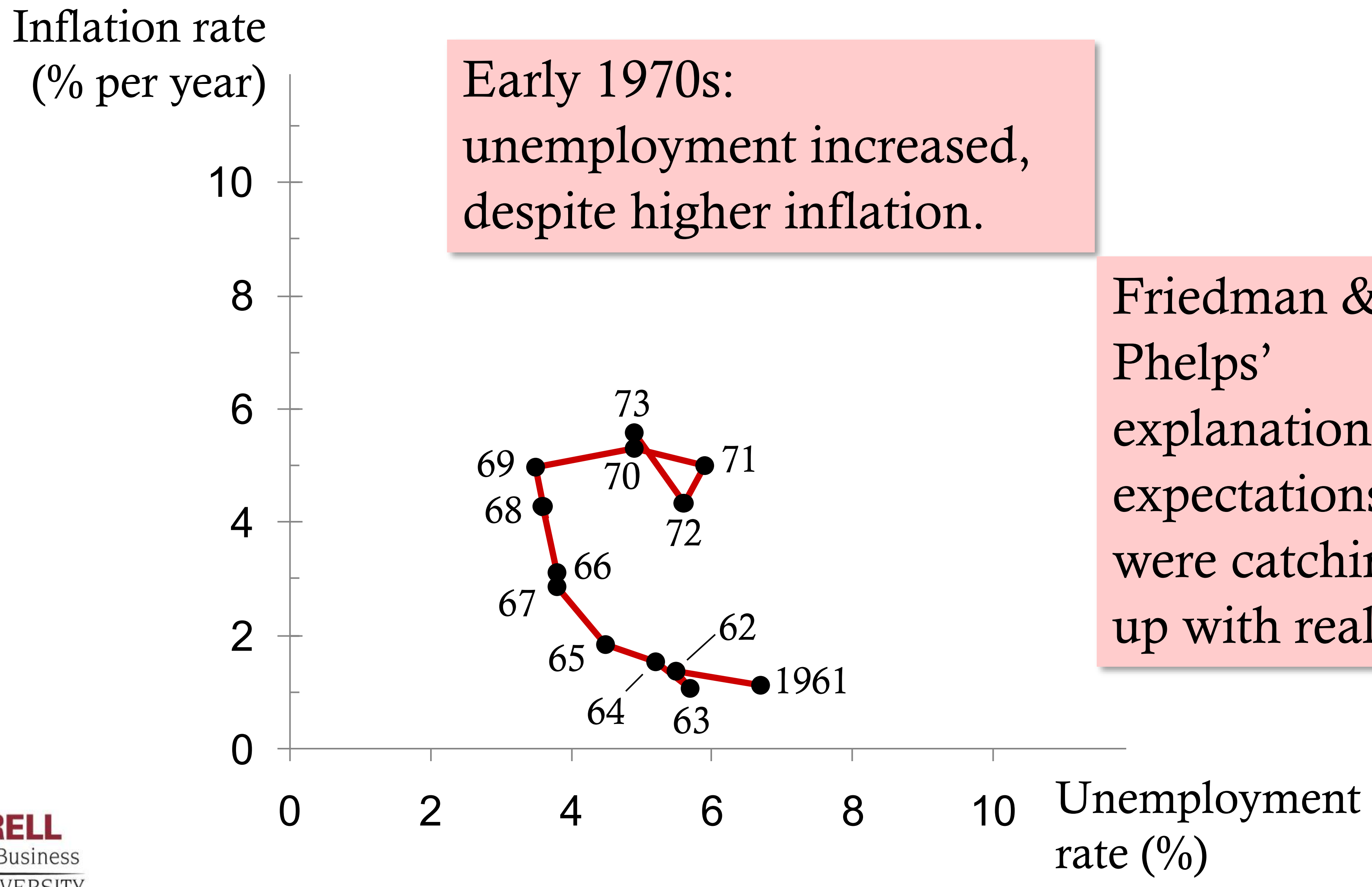
A. Plot the long-run Phillips curve.
B. Find the u-rate for each of these values of actual inflation: 0%, 6%. Sketch the short-run *PC*.
C. Suppose expected inflation rises to 4%. Repeat part B.
D. Instead, suppose the natural rate falls to 4%. Draw the new long-run Phillips curve, then repeat part B.

# Activity: Shifting the Phillips Curve

- An increase in expected inflation shifts *PC* to the right.
- A fall in the natural rate shifts both curves to the left.



# The Breakdown of the Phillips Curve




Early 1970s:
unemployment increased,
despite higher inflation.

Friedman & Phelps' explanation: expectations were catching up with reality.

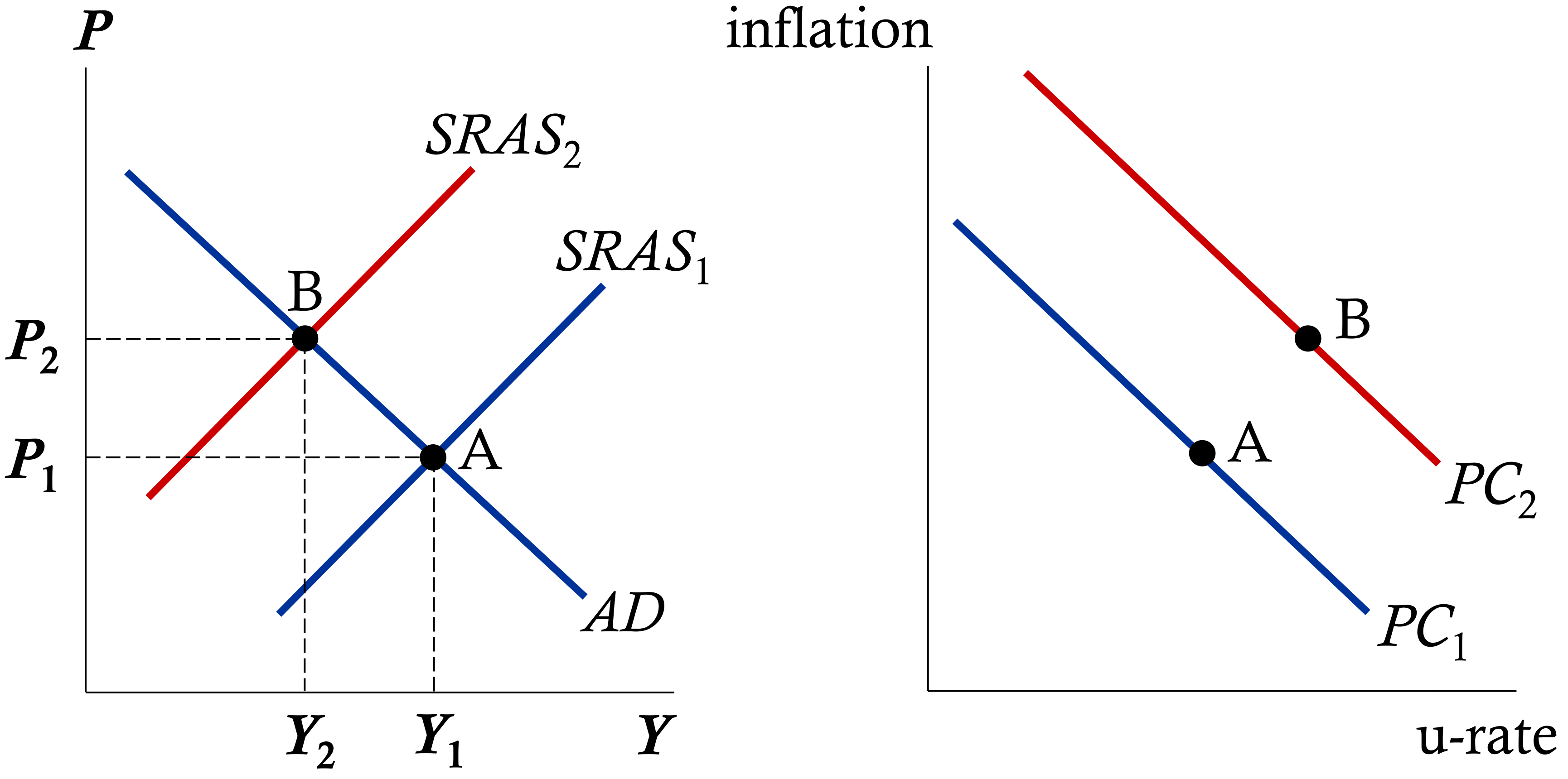
# Another Phillips Curve Shifter: Supply Shocks

- **Supply shock**:
  an event that directly alters firms' costs and prices, shifting the *AS* and *PC* curves

- Example:  large increase in oil prices

# How an adverse supply shock shifts the PC

*SRAS* shifts left, prices rise, output & employment fall.



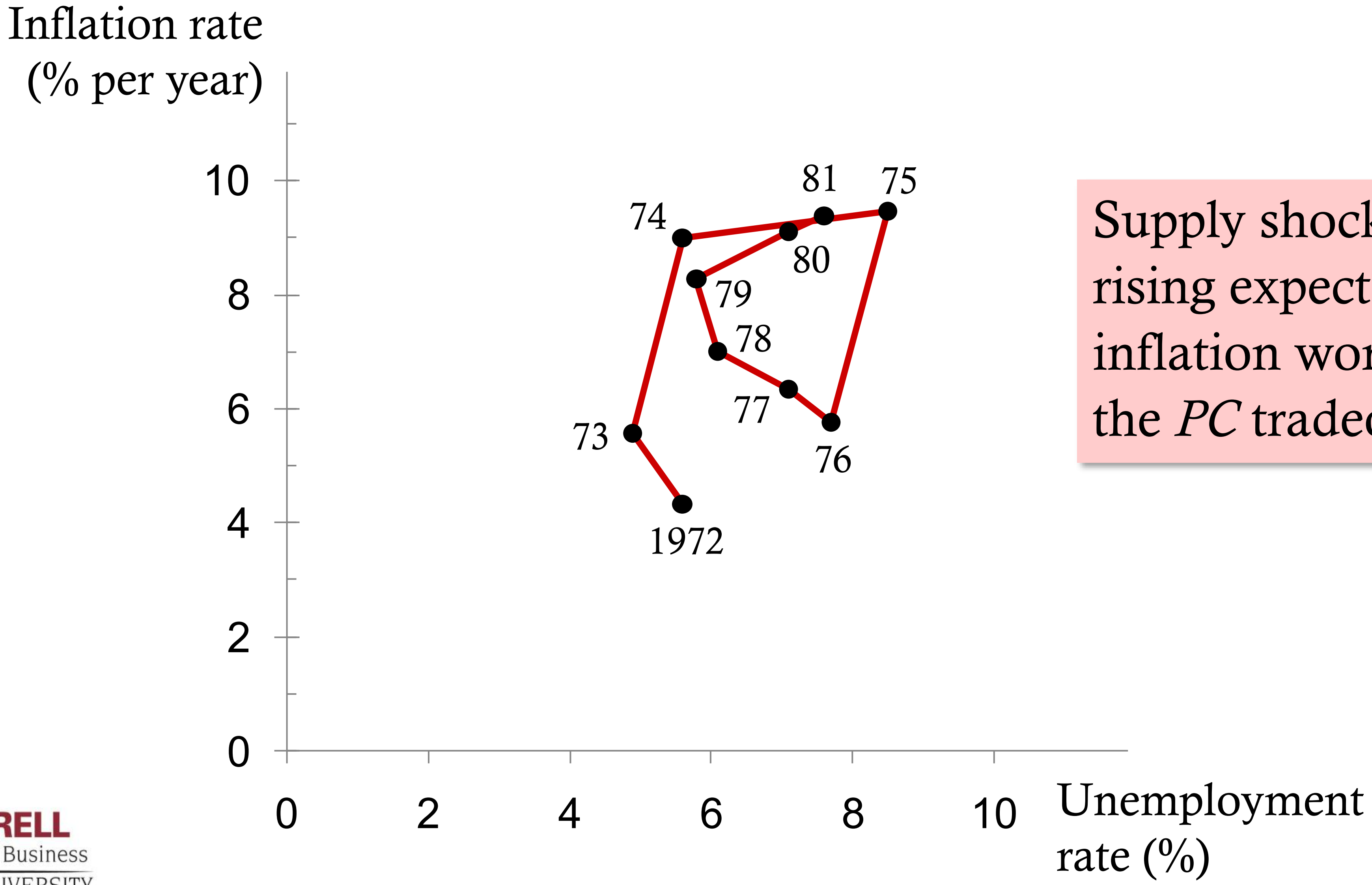

Inflation & u-rate both increase as the *PC* shifts upward.

# Supply Shock: 1970's Oil Prices

| Oil price per barrel | |
|---|---|
| 1/1973 | $ 3.56 |
| 1/1974 | 10.11 |
| 1/1979 | 14.85 |
| 1/1980 | 32.50 |
| 1/1981 | 38.00 |

- The Fed chose to accommodate the first shock in 1973 with faster money growth.
- Result: Higher expected inflation, which further shifted *PC*.
- 1979: Oil prices surged again, worsening the Fed's tradeoff.

# Supply Shock: 1970's Oil Prices




Supply shocks & rising expected inflation worsened the *PC* tradeoff.

# Activity: Understanding a Supply Shock and Its Effects on the Phillips Curve

1. A sudden spike in global oil prices raises firms' production costs across the economy.
   Using the Phillips curve framework, explain why both inflation and unemployment increase in the short run.
   In your answer, identify whether this is a movement along the existing SRPC or a shift of the SRPC.

2. After the supply shock, suppose policymakers choose to accommodate the higher inflation with expansionary monetary policy to prevent unemployment from rising.
   Describe how this policy affects expected inflation and what happens to the short-run Phillips curve over time.

# Activity: Understanding a Supply Shock and Its Effects on the Phillips Curve

- A spike in oil prices raises production costs, shifting the short-run aggregate supply (SRAS) curve left. This leads to higher inflation and lower output, which means higher unemployment.
- In the Phillips curve framework, this is not a movement along the SRPC. Instead, it causes the short-run Phillips curve to shift upward because at every unemployment rate, inflation is now higher due to increased costs.

# Activity: Understanding a Supply Shock and Its Effects on the Phillips Curve

- If policymakers accommodate the supply shock (e.g., by increasing money growth), they prevent unemployment from rising—but at the cost of higher inflation.
- Over time, households and firms begin to expect this higher inflation. As expected inflation rises, the short-run Phillips curve shifts further upward.
  The economy ends up with permanently higher inflation, and unemployment eventually returns to the natural rate.

# The Cost of Reducing Inflation

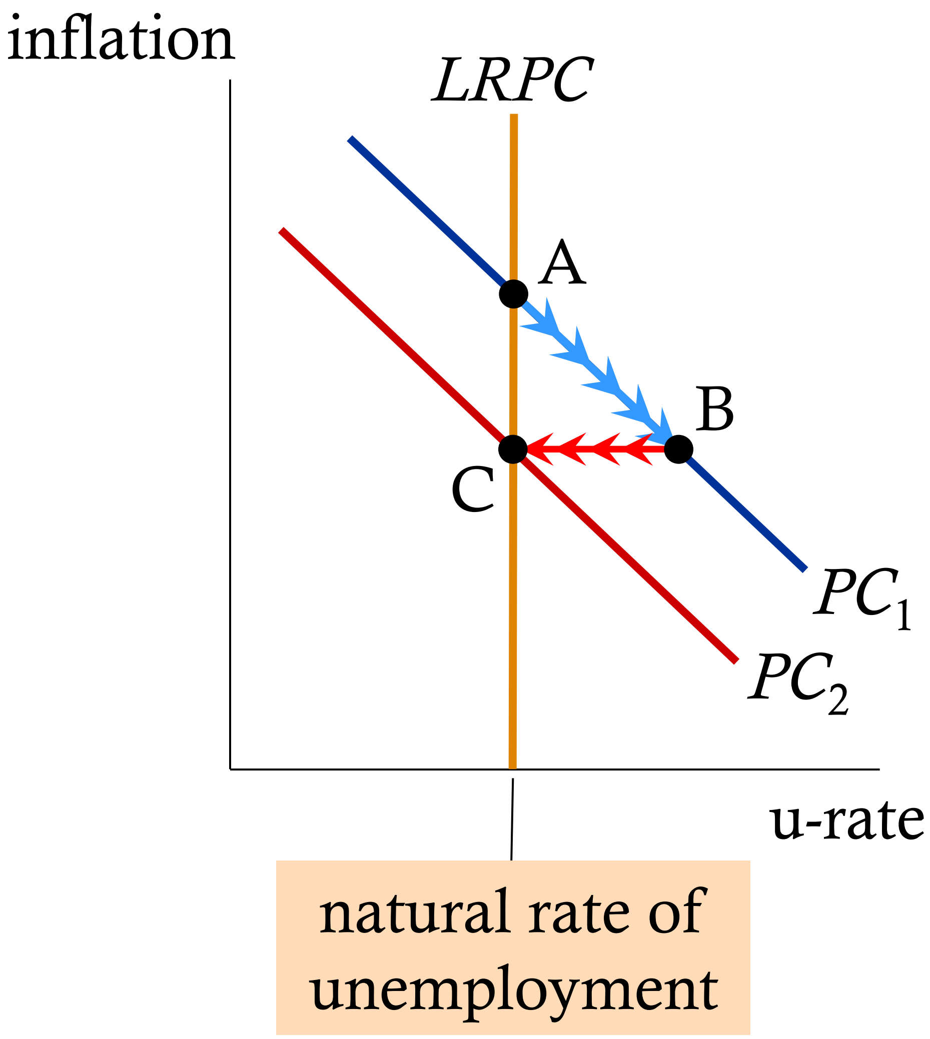
- **Disinflation**:  a reduction in the inflation rate
- To reduce inflation, Fed must slow the rate of money growth, which reduces aggregate demand.
- Short run: Output falls and unemployment rises.
- Long run: Output & unemployment return to their natural rates.

# Disinflationary Monetary Policy

Contractionary monetary policy moves economy from A to B.

Over time, expected inflation falls, *PC* shifts downward.

In the long run, point C: the natural rate of unemployment, lower inflation.



# The Cost of Reducing Inflation

- Disinflation requires enduring a period of high unemployment and low output.
- **Sacrifice ratio**:
  percentage points of annual output lost per 1 percentage point reduction in inflation
- Typical estimate of the sacrifice ratio: 5
  - To reduce inflation rate 1%, must sacrifice 5% of a year's output.
- Can spread cost over time, e.g. to reduce inflation by 6%, can either
  - sacrifice 30% of GDP for one year
  - sacrifice 10% of GDP for three years

# Rational Expectations, Costless Disinflation?

- **Rational expectations**: a theory according to which people optimally use all the information they have, including info about government policies, when forecasting the future

- Early proponents:
  Robert Lucas, Thomas Sargent, Robert Barro

- Implied that disinflation could be much less costly…

- Tick, tock, tick, tock, tick, ______…

# Rational Expectations, Costless Disinflation?

- Suppose the Fed convinces everyone it is committed to reducing inflation.

- Then, expected inflation falls, the short-run *PC* shifts downward.

- Result: Disinflation can cause less unemployment than the traditional sacrifice ratio predicts.

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# The Volcker Disinflation

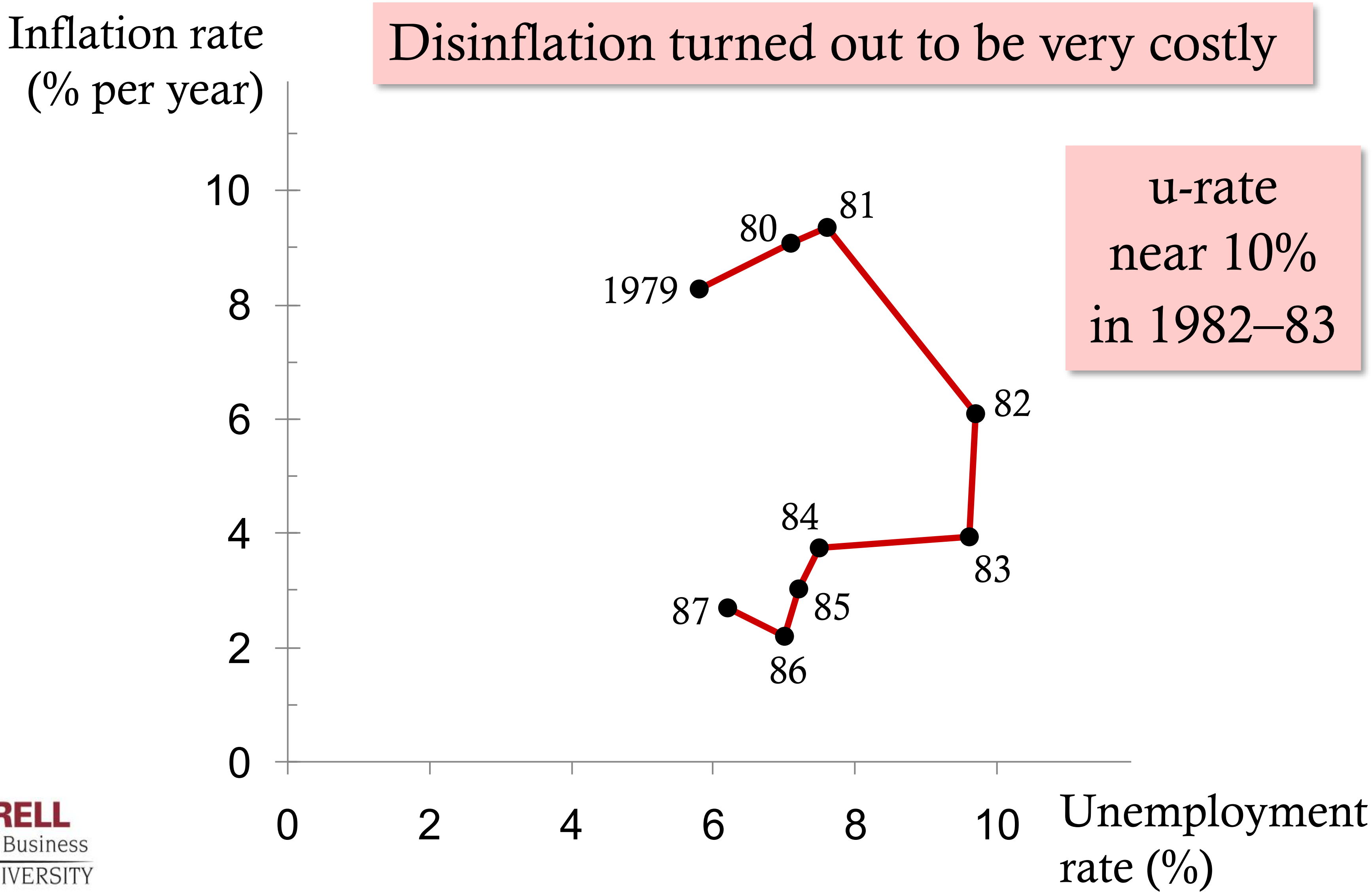
Fed Chairman Paul Volcker

- Appointed in late 1979 under high inflation & unemployment
- Changed Fed policy to disinflation

1981–1984:

- Fiscal policy was expansionary, so Fed policy had to be <u>very</u> contractionary to reduce inflation.
- Success: Inflation fell from 10% to 4%,but at the cost of high unemployment…

# The Volcker Disinflation

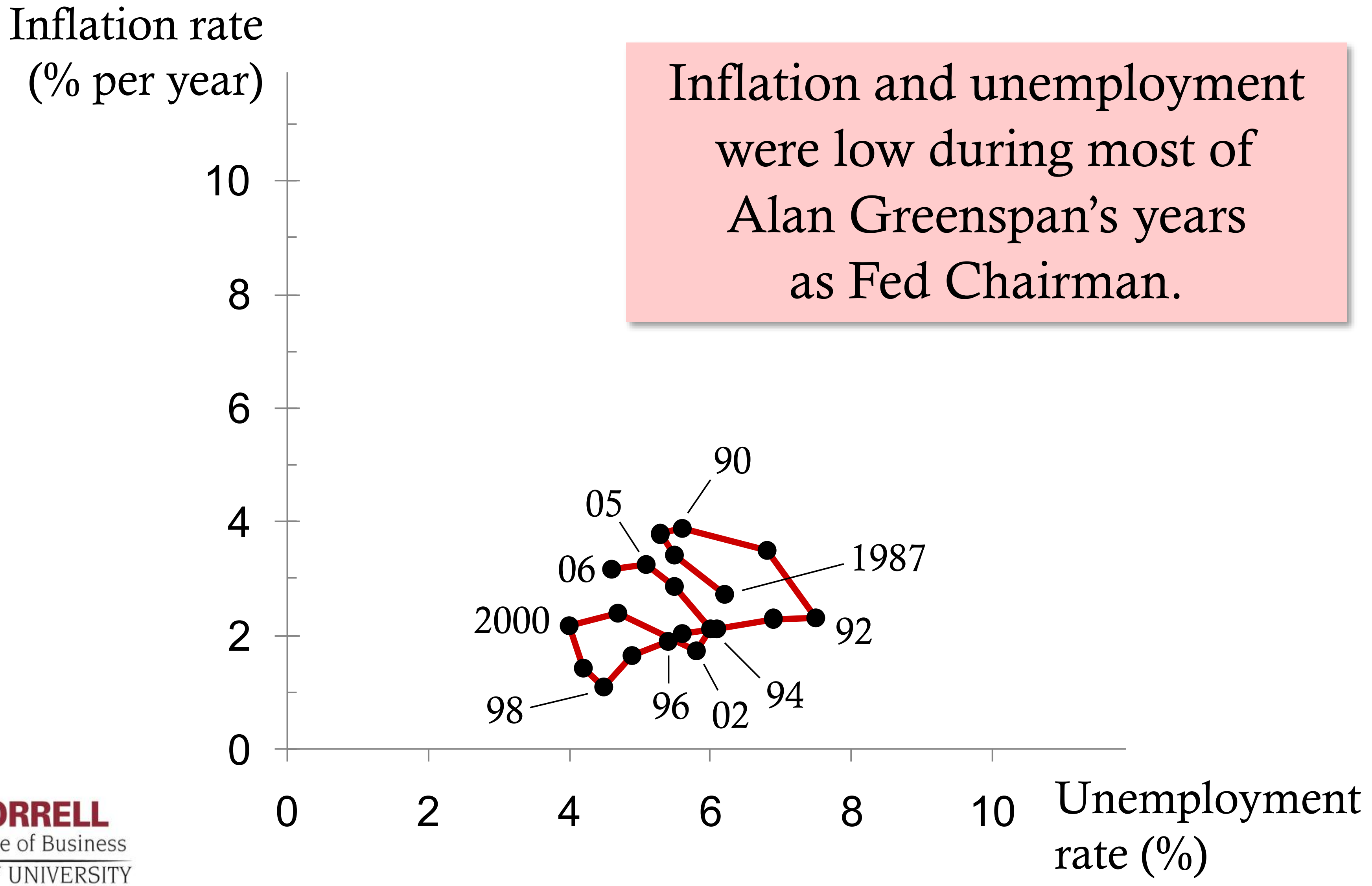


# The Greenspan Era

- 1986: Oil prices fell 50%.
- 1989–90: Unemployment fell, inflation rose. Fed raised interest rates, caused a mild recession.
- 1990s: Unemployment and inflation fell.
- 2001: Negative demand shocks created the first recession in a decade. Policymakers responded with expansionary monetary and fiscal policy.

# The Greenspan Era

Inflation and unemployment were low during most of Alan Greenspan's years as Fed Chairman.

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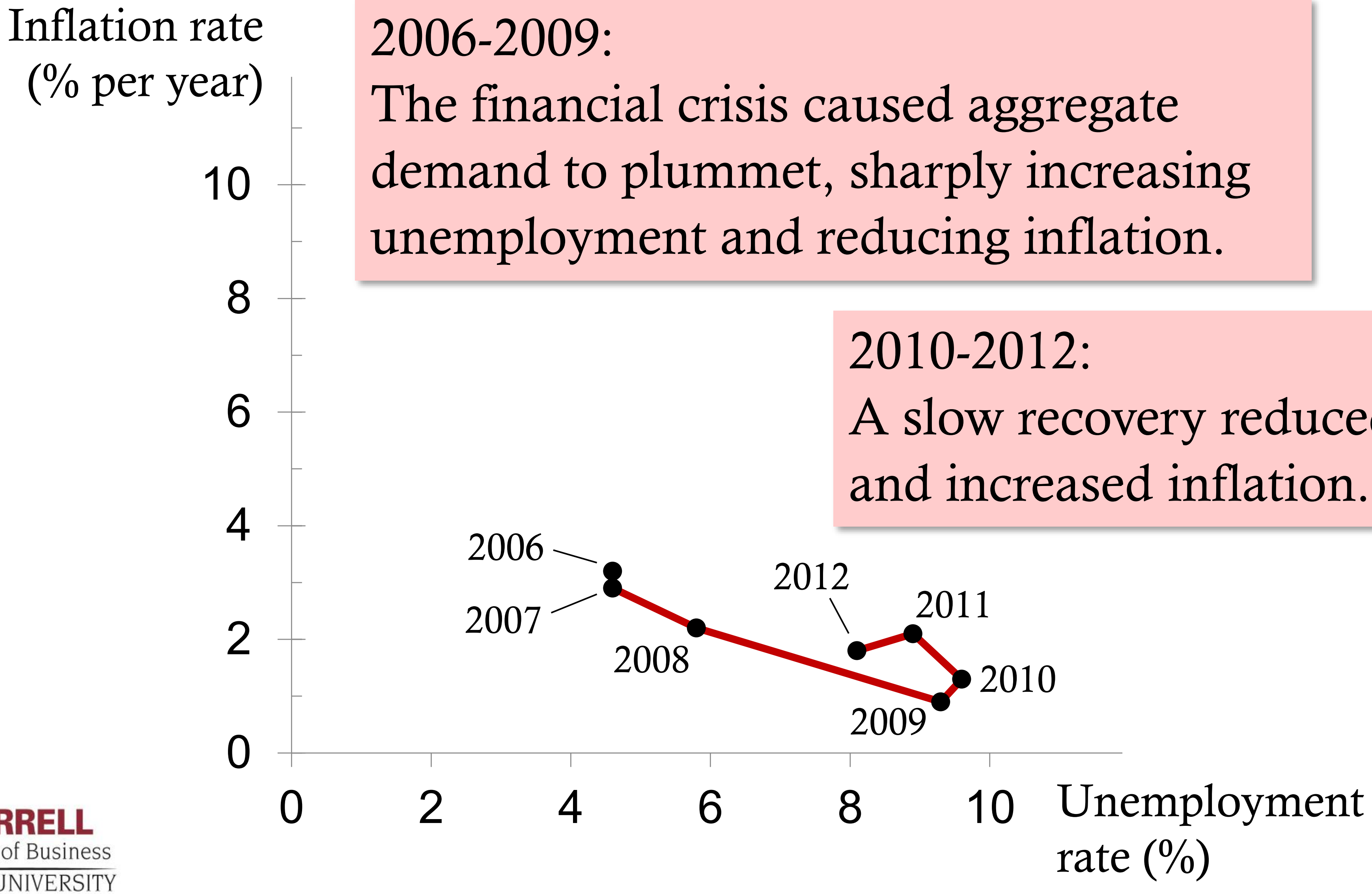
# The Phillips Curve During the Financial Crisis

- The early 2000s housing market boom turned to bust in 2006
- Household wealth fell, millions of mortgage defaults and foreclosures, heavy losses at financial institutions
- Result:
  Sharp drop in aggregate demand, steep rise in unemployment

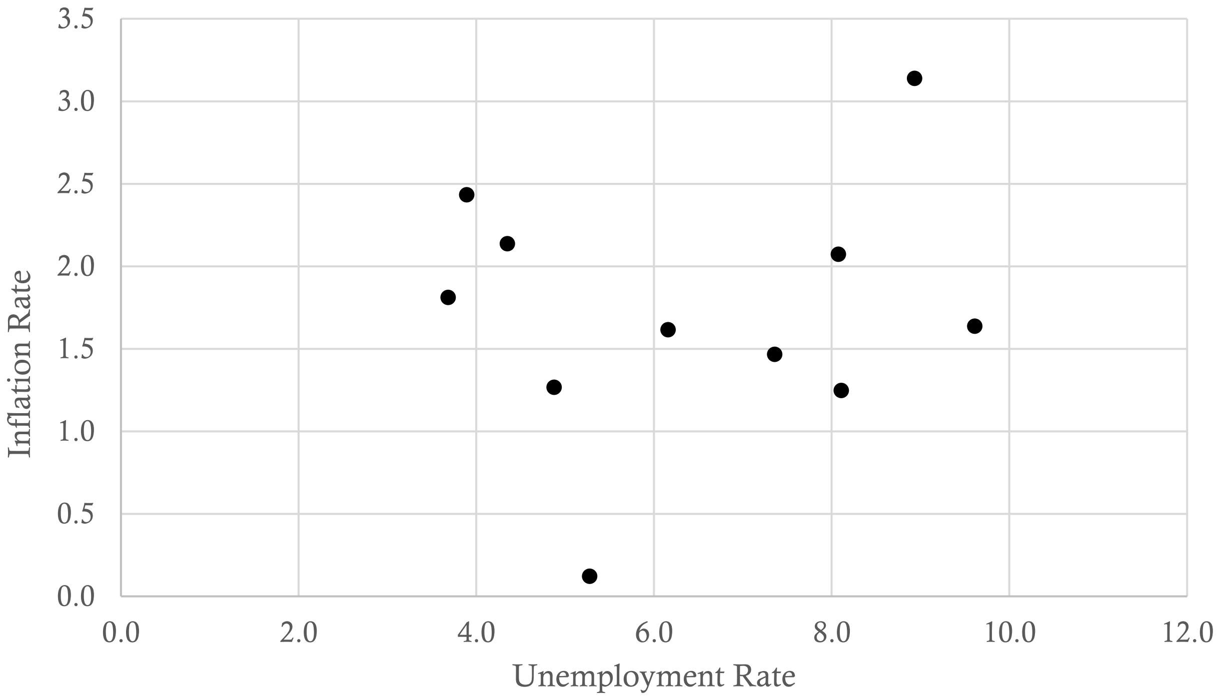
# The Phillips Curve During the Financial Crisis




**2006-2009:**
The financial crisis caused aggregate demand to plummet, sharply increasing unemployment and reducing inflation.

**2010-2012:**
A slow recovery reduced unemployment and increased inflation.

# Return to Normal?




| | Inflation Rate | Unemployment Rate |
|---|---|---|
| 2010 | 1.6 | 9.6 |
| 2011 | 3.1 | 8.9 |
| 2012 | 2.1 | 8.1 |
| 2013 | 1.5 | 7.4 |
| 2014 | 1.6 | 6.2 |
| 2015 | 0.1 | 5.3 |
| 2016 | 1.3 | 4.9 |
| 2017 | 2.1 | 4.4 |
| 2018 | 2.4 | 3.9 |
| 2019 | 1.8 | 3.7 |
| 2020 | 1.2 | 8.1 |

# US Inflation and Unemployment 2020-2025

**Overview of the Period**

- In 2020, the COVID-19 pandemic and associated lockdowns triggered a massive rise in unemployment (peaking around 14.8 % in April 2020).
- Inflation was subdued in early 2020 but then surged, peaking in 2022 around 9 %.
- From late 2022 through early 2025, inflation declined by roughly 5 percentage points without a large increase in unemployment.
- As of mid-2025, unemployment was around 4.3 %.

# US Inflation and Unemployment 2020-2025

**Key Features Relevant to the Phillips Curve**

- Very tight labor market (2021-22): High vacancy-to-unemployment ratios likely raised inflationary pressure.
- Supply shocks: e.g., global energy/commodity price spikes, disrupted supply chains after the pandemic. These likely shifted the short-run Phillips curve (SRPC) upward.
- Expectations & anchoring: Even as inflation fell, unemployment didn't spike dramatically, suggesting shifts in the Phillips curve rather than simply movements along it.
- Policy response: The Federal Reserve and fiscal policies played strong roles; accommodating recovery then tightening as inflation surged.

# US Inflation and Unemployment 2020-2025

**Interpretation: What Does This Case Study Tell Us?**

- SR trade-off is messy: The classic negative trade-off (lower unemployment → higher inflation) is present but distorted by major shocks.
- Curve shifts matter: The data suggest the SRPC moved upward (worse trade-off) during 2021-22 and may have shifted again during the disinflation phase.
- Natural rate and long-run: Despite falling inflation from 2022-25, unemployment did not fall dramatically below its natural rate, reinforcing the idea of the long-run Phillips curve being vertical / less trade-off.
- Expectations and structural change: The relative stability of unemployment during disinflation indicates expectations and structural factors (e.g., labor supply) played important roles.

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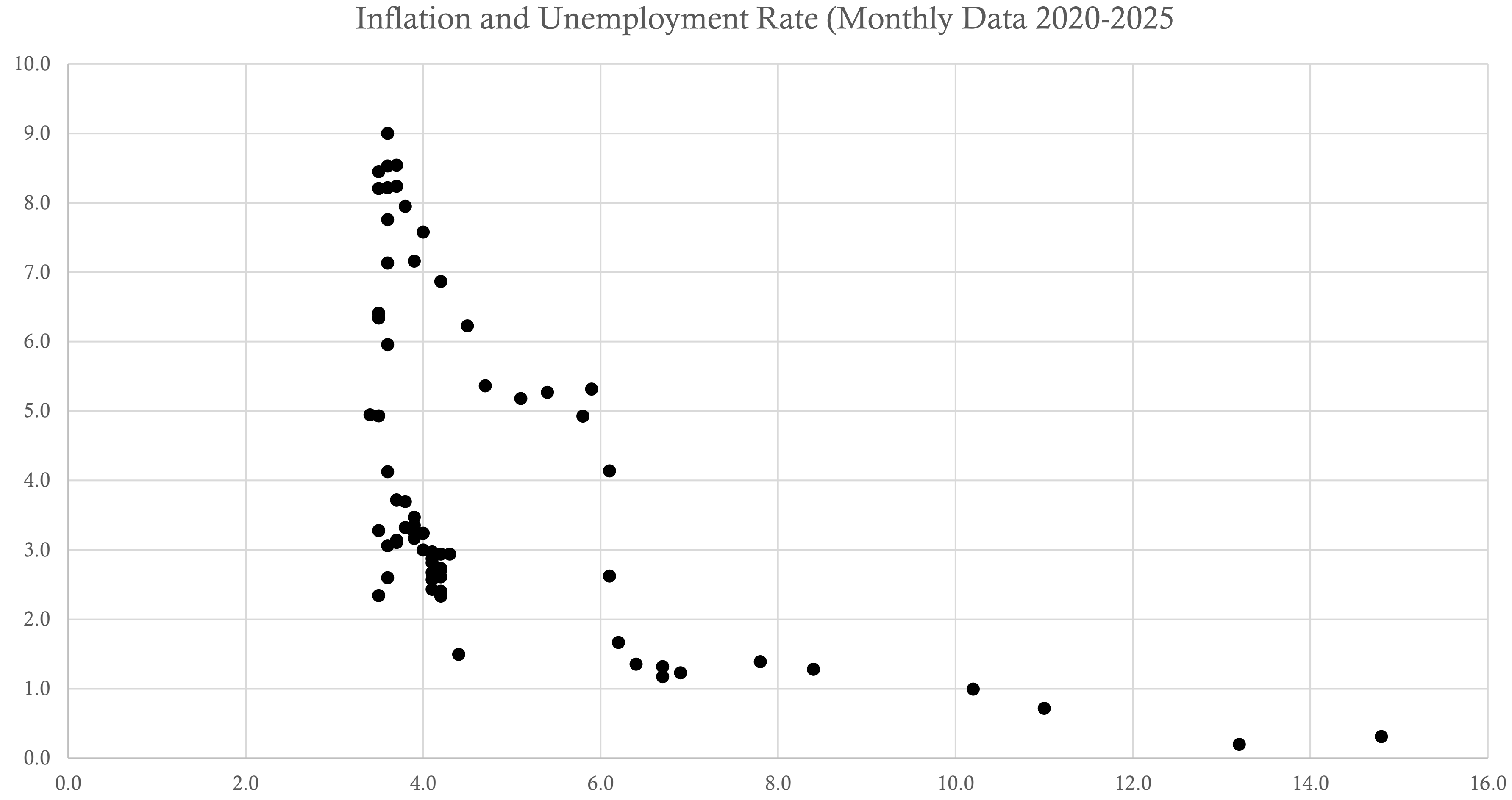
# US Inflation and Unemployment 2020-2025

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# Key Takeaways

- The theories in this chapter come from some of the greatest economists of the 20th century.
- They teach us that inflation and unemployment are
  - unrelated in the long run
  - negatively related in the short run
  - affected by expectations,
    which play an important role in the economy's adjustment from the short-run to the long run

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# Knowledge Check

1. Why is the long-run Phillips curve vertical?
2. What role do expectations play in shifting the Phillips curve?
3. What happens to the short-run Phillips curve if expected inflation increases?
4. When the Fed contracts the money supply to reduce inflation, what happens to unemployment in the short run?
5. Why was the Volcker disinflation costly?

# Knowledge Check Answers

1. Vertical long-run Phillips curve: in the long run, unemployment returns to its natural rate regardless of inflation.
2. Expectations: Higher expected inflation shifts the short-run Phillips curve upward/right.
3. Rising expected inflation: the short-run Phillips curve shifts upward.
4. Contractionary monetary policy: reduces inflation but temporarily raises unemployment.
5. Volcker disinflation cost: very tight monetary policy caused prolonged high unemployment.